

# C2G Parameter Golf Challenge

Technical Report: Extreme-Constraint Language Model Training on a 10-Minute / 16 MB Budget

|                     |                                                                                  |
|---------------------|----------------------------------------------------------------------------------|
| Submitter           | lenovo                                                                           |
| Track               | track_10min_16mb (10-minute wall clock, artifact <= 16,000,000 bytes)            |
| Date                | 2026-10-04                                                                       |
| Target level        | Level 4 (val_bpb < 1.085), design point 1.0810                                   |
| Seeds               | 0 / 42 / 1234                                                                    |
| Reference SOTA      | bigbag, 2026-04-09_SP8192_3LayerRecur_ParResid_QK525_LegalTTT, val_bpb = 1.08100 |
| Reference baseline  | Baseline, 2026-03-17_NaiveBaseline, val_bpb = 1.2243657                          |
| Verification status | PENDING_VERIFICATION                                                             |

## Abstract

This report documents a parameter-golf submission for the C2G challenge: training a language model from scratch within a 600-second wall-clock budget on 8xH100-80GB, then quantizing and packaging the resulting weights into an artifact no larger than 16,000,000 bytes, and finally evaluating bits-per-byte (val\_bpb) under the official evaluation protocol. The proposed recipe combines a 8192-token SentencePiece vocabulary (SP8192), an 11-layer / 512-dimension decoder with 3-layer depth recurrence, parallel residuals from layer 7 onward, the MuonEq-R matrix optimizer with a high weight decay of 0.095, a fixed QK gain of 5.25, EMA (0.9965), Hessian-guided SDClip with All-Int6 GPTQ quantization (6-bit matrices at 12.85 sigma; 8-bit embeddings at 20.0 sigma), Brotli-11 container compression, sliding-window evaluation, and a compliance-clean Legal Score-First test-time-training (TTT) pass. Every performance number quoted in this report is taken from the public record table of the official repository. No training run was executed on the machine that produced this report; see Section 7 for the mandatory disclosure.

## Verification status notice (read first)

The val\_bpb of 1.0810 attributed to this submission is a TARGET value, not a measured result. The authoring machine has no 8xH100 node (0 H100 GPUs present); consequently no training, no quantization and no evaluation were executed here. In the machine-readable submission record (lenovo\_C2G\_submission.json) the field "verification\_status" is set to "pending\_verification" and every field that cannot be measured without a real run (val\_bpb\_std, seed\_results, artifact\_bytes, the timing-related compliance flags and three\_seeds) is set to null. The three seed log files under lenovo\_C2G\_logs/ are explicitly labelled "LOG STATUS: PLANNED / NOT EXECUTED" and every metric line in them carries a [TARGET] prefix.

# 1. Problem Statement and Constraints

The C2G parameter-golf track is a compute-constrained optimization problem with four hard constraints that bind simultaneously. First, the total training wall clock must not exceed 600 seconds on the reference hardware. Second, the serialized artifact must not exceed 16,000,000 bytes; the artifact is measured after quantization and after container compression, so the weight precision budget and the compression choice are part of the modeling problem, not a post-processing detail. Third, the evaluation pass must fit inside the same 600-second budget. Fourth, submissions are grouped by seed and the headline number is the mean over three independent seeds, so run-to-run stability matters as much as the peak score.

The optimization target is validation bits-per-byte on the official held-out split: lower is better. The naive baseline of the official repository sits at 1.2243657 BPB and the current public state of the art sits at 1.08100 BPB (std 0.00020). Closing this 0.1434 BPB gap within 600 seconds requires every component in the recipe to pay for itself; a component that is merely neutral is a loss, because it consumes time that a paid component could have used.

## 1.1 Why the budget forces a co-design

Under a fixed parameter-byte budget, three levers exist: (i) increase tokens per parameter by shrinking the model and re-using layers, (ii) reduce the bytes consumed per parameter via quantization, and (iii) recover the quantization loss at evaluation time via test-time training. The submitted design uses all three. Depth recurrence re-uses physical layers 3 to 5 to raise effective depth without raising parameter count; All-Int6 GPTQ plus Brotli-11 buys roughly a 2.6x reduction against fp16 storage; and Legal Score-First TTT recovers part of the quantization error by adapting on the scored sequence itself, in a way that does not violate the compliance rules.

## 2. Evidence Base: Public Records Used for Calibration

All reference values below are copied verbatim from records/\*/submission.json of the official repository. They are reproduced here so that any reader can re-derive the design choices in Section 3 from primary sources.

| Record directory                                             | Author       | val_bpb    | Artifact bytes                       |
|--------------------------------------------------------------|--------------|------------|--------------------------------------|
| 2026-03-17_NaiveBaseline                                     | Baseline     | 1.2243657  | 15,863,489                           |
| 2026-03-19_SlidingWindowEval                                 | mattqlf      | 1.19250007 | -                                    |
| 2026-03-31_ParallelResiduals_MiniDepthRecurrence             | Robby955     | 1.10625353 | -                                    |
| 2026-04-03_MuonEqR_DepthRecurrence_WD090_AllInt6             | dexhunter    | 1.0912     | -                                    |
| 2026-04-04_SP4096_...                                        | aryanbhosale | 1.08971631 | -                                    |
| 2026-04-05_SP8192_GPTQ-Embeddings_SDClip_Loop45x2            | clarkkev     | 1.08563    | -                                    |
| 2026-04-06_SP8192_QK5_LegalTTT_1.0828                        | dexhunter    | 1.08279384 | 15,991,018 / 15,992,546 / 15,989,058 |
| 2026-04-06_SP8192_HessianSDClip_ProgressiveRecurrence        | Robby955     | 1.08354    | -                                    |
| 2026-04-08_...                                               | bigbag       | 1.08217956 | -                                    |
| 2026-04-08_SP8192_...                                        | aryanbhosale | 1.08218    | -                                    |
| 2026-04-09_SP8192_3LayerRecur_ParResid_QK525_LegalTTT (SOTA) | bigbag       | 1.08100    | 15,991,930 / 15,992,919 / 15,993,232 |

Note on seed sets. The current SOTA record reports seeds 42 / 314 / 999 with a standard deviation of 0.00020; the record immediately below it (dexhunter, 1.08279384) uses seeds 0 / 42 / 1234. This submission adopts 0 / 42 / 1234 so that its result is directly comparable to the same-seed reference rather than to a differently-seeded one.

## 3. Method

### 3.1 Model architecture

The backbone is an 11-layer decoder-only Transformer with 512 embedding dimensions, 8 attention heads and 4 key/value heads (grouped-query attention), a 4x MLP expansion and tied input/output embeddings. Three architectural modifications carry the parameter-efficiency argument:

**Depth recurrence.** Physical layers 3, 4 and 5 are executed a second time, so the network behaves like a 14-layer stack while storing only 11 layers. Recurrence is switched on only after 35% of the training progress has elapsed (`ENABLE_LOOPING_AT` = 0.35) so that the non-recurrent pathway is learned first; the loop is limited to a contiguous middle block (L3-L5) to avoid gradient interference at the input and output ends.

**Parallel residuals.** From layer 7 onward the attention and MLP sub-layers receive the same input and their outputs are summed, instead of being chained. This shortens the gradient path in the deep half of the stack, which matters because the recurrent block doubles the effective depth of that region. The first six layers remain serial to preserve low-level feature extraction.

**QK gain.** The query-key dot product is multiplied by a fixed gain of 5.25. Larger gains sharpen the attention distribution, which compensates for the softened logits introduced by 6-bit matrix quantization. The value is taken from the current SOTA record and is not re-tuned here.

### 3.2 Optimization

Matrix parameters (attention and MLP projections) are updated with MuonEq-R under row-normalized momentum with a weight decay of 0.095; scalar and embedding parameters use AdamW at the same decay. The learning-rate schedule is a 250-step warmup followed by a cosine-to-floor decay; QAT (quantization-aware training) with a straight-through estimator on the int6 matrices begins at approximately step 350. An exponential moving average with decay 0.9965 supplies the weights that are ultimately quantized, which both smooths late-training noise and improves the post-quantization score. The training script stops early at approximately step 4550, mirroring the stop point observed in the SOTA record, because the marginal BPB gain per step after that point no longer justifies the wall-clock risk.

### 3.3 Hyperparameters

| Group        | Parameter                                        | Value                           |
|--------------|--------------------------------------------------|---------------------------------|
| Tokenizer    | type / vocab                                     | SP8192 / 8192                   |
| Architecture | layers / d_model / heads / KV heads              | 11 / 512 / 8 / 4                |
| Architecture | MLP multiplier / tied embeddings                 | 4x / True                       |
| Architecture | train seq len / eval seq len / eval stride       | 2048 / 2048 / 64                |
| Recurrence   | enable_looping_at / num_loops / loop layer block | 0.35 / 2 / L3-L5                |
| Residuals    | parallel_residual_start                          | 7                               |
| Attention    | qk_gain_init / partial RoPE dim                  | 5.25 / 16                       |
| Optimizer    | matrix_lr / scalar_lr / tied_embed_lr            | 0.025 / 0.025 / 0.035           |
| Optimizer    | muon WD / adam WD / muon momentum / row norm     | 0.095 / 0.095 / 0.99 / True     |
| Schedule     | warmup steps / warmup fraction / EMA decay       | 250 / 0.15 / 0.9965             |
| Quantization | matrix bits / matrix clip sigma                  | 6 / 12.85                       |
| Quantization | embed bits / embed clip sigma                    | 8 / 20.0                        |
| Quantization | container / pre-quant TTT                        | Brotli-11 / disabled            |
| TTT          | lr / epochs / chunk / order                      | 0.005 / 3 / 32768 / score-first |
| Runtime      | seeds / wall-clock budget                        | 0, 42, 1234 / 600 s             |

## 4. Quantization, Packaging and Test-Time Training

### 4.1 Quantization pipeline

The quantization pipeline runs strictly after training and strictly before packaging. The order is load-bearing: the Hessian used by SDClip is computed from the trained model, the GPTQ rounding decisions are made per layer against that Hessian, and the embedding matrix is treated separately because it dominates the artifact under a large vocabulary.

| Stage | Operation                  | Setting                              |
|-------|----------------------------|--------------------------------------|
| 1     | Freeze EMA weights         | decay 0.9965                         |
| 2     | Collect Hessian statistics | calibration split, per linear layer  |
| 3     | SDClip threshold search    | matrix 12.85 sigma, embed 20.0 sigma |
| 4     | GPTQ layer-wise rounding   | matrix int6, embed int8              |
| 5     | Container compression      | Brotli level 11                      |
| 6     | Size gate                  | assert bytes &lt;= 16,000,000        |

The size gate is a hard assertion, not a warning. If the packed artifact exceeds 16,000,000 bytes the run is invalid regardless of BPB, so the design deliberately targets approximately 15,990,000 bytes to leave a safety margin of about 10,000 bytes for container overhead variation.

### 4.2 Why embeddings stay at 8 bits

With an 8192-token vocabulary and tied embeddings, the embedding matrix holds 8192 x 512 values, which is a large share of the total parameter count. Quantizing it to 6 bits would free budget but measurably degrades BPB, because embedding precision acts as input noise on every token. The public records bracket this trade-off: an All-Int6 configuration lands at 1.0912 while a configuration that keeps 8-bit embeddings lands at 1.08279384 under the same 16 MB ceiling. This submission therefore spends its residual byte budget on embedding precision.

### 4.3 Legal Score-First TTT

Test-time training adapts the quantized model on the evaluation sequence before scoring it. The score-first ordering means the scoring pass is executed in an order that never lets the model see the token it is about to predict. Four compliance switches are held explicitly off: no slot mechanism, no pre-quantization TTT, no eval-time layer banking and no n-gram cache. TTT runs with SGD at learning rate 0.005 for 3 epochs over chunks of 32768 tokens, and it sees the quantized weights so that it can compensate for quantization error rather than for training error. This is the single most time-expensive evaluation component and is budgeted separately inside the same 600-second ceiling.

## 5. Ablation Design

Public records form a date chain in which several components are changed between consecutive entries, which makes causal attribution impossible from the records alone. This submission therefore defines a single-knob ablation matrix. Each row changes exactly one component relative to the full configuration A0, with all other settings frozen. The decision rule is a difference of at least 0.0006 BPB, corresponding to roughly three times the observed seed standard deviation of 0.0002, so that noise is not mistaken for signal.

| ID | Configuration                       | Knob changed vs A0                 | Status               |
|----|-------------------------------------|------------------------------------|----------------------|
| A0 | Full recipe                         | - (reference)                      | pending_verification |
| A1 | A0 minus depth recurrence           | num_loops 2 -&gt; 1                | pending_verification |
| A2 | A0 with serial residuals everywhere | parallel_residual_start 7 -&gt; 11 | pending_verification |
| A3 | A0 with lower QK gain               | qk_gain 5.25 -&gt; 5.00            | pending_verification |
| A4 | A0 with all-6-bit embeddings        | embed_bits 8 -&gt; 6               | pending_verification |
| A5 | A0 with plain Muon (no Eq-R)        | muon_row_norm True -&gt; False     | pending_verification |
| A6 | A0 without TTT                      | ttt_epochs 3 -&gt; 0               | pending_verification |
| A7 | A0 with single-window eval          | eval_stride 64 -&gt; 2048          | pending_verification |
| A8 | A0 without QAT warmdown             | qat_start -&gt; disabled           | pending_verification |

Two interaction terms are also defined, because the components are not independent: recurrence interacts with parallel residuals (both change the effective gradient path) and quantization precision interacts with TTT (TTT can absorb part of the quantization error, which reduces the observed penalty of aggressive quantization). These interactions are reported in lenovo\_C2G\_ablation.md; none of the rows above have been executed, and all carry the pending\_verification status.

## 6. Compliance Statement

| Compliance item     | This submission      | Basis                                             |
|---------------------|----------------------|---------------------------------------------------|
| train_under_600s    | pending_verification | target 588 s; not executed                        |
| artifact_under_16mb | pending_verification | target ~15,990,000 B; model artifact not produced |
| eval_under_600s     | pending_verification | not executed                                      |
| no_slot             | TRUE                 | design holds the slot mechanism off               |
| no_pre_quant_ttt    | TRUE                 | TTT runs only on quantized weights                |
| no_etlb             | TRUE                 | eval-time layer banking disabled                  |
| no_ngram_cache      | TRUE                 | no n-gram cache in the evaluation path            |
| score_first_ttt     | TRUE                 | score-first sampling order enforced               |
| three_seeds         | pending_verification | seeds 0 / 42 / 1234 planned                       |

**The four items marked TRUE are design properties that can be verified by reading the training script. The five items marked pending\_verification are measurement claims that require an actual run on the reference hardware. They are deliberately not asserted here. Marking an unmeasured property as satisfied would invalidate the submission under the honesty requirement, which is stricter than the performance requirement.**

## 7. Verification Status and Limitations

This section is the mandatory disclosure that accompanies the artifact. It exists because the distinction between a plan and a result is the difference between a scientific claim and a fabrication.

| Limitation                                                          | Consequence                                                                                                                             | How it is handled                                                       |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| No 8xH100 node available (0 H100 GPUs on the training organization) | Training, organization or evaluation could be excluded from the submission.                                                             | SBP is a target; unmeasurable                                           |
| Seed logs are not produced by real runs                             | The logs cannot be used as evidence of convergence.                                                                                     | Each log file is headed 'LOG STATUS: PLANNED / NOT EXECUTED'            |
| The packaged .tar.gz is a reproduction bundle                       | The artifact bytes and weights reported as a measured artifact bytes is null; artifact_bytes_target = 15,990,000 is reported separately |                                                                         |
| Hyperparameters are reused from public records                      | The results may be suboptimal for the chosen seed set                                                                                   | Seed set 0/42/1234 was chosen to match an existing same-seed public     |
| Ablation rows are defined but not executed                          | No causal component attribution can be claimed                                                                                          | Every ablation row is marked pending_verification and no effect size is |

What can be verified today, without a GPU, is the internal consistency of the submission package: every reference number in this report and in the accompanying documents traces back to a records\*/submission.json file; the seed set, the artifact byte targets and the compliance flags agree across the submission record, the README, the ablation table and the leaderboard file; and the packaged bundle is a real, openable archive below the size limit.

## 8. Reproduction Procedure

The procedure below is intended to be executed exactly as written on a node with 8xH100-80GB SXM GPUs, PyTorch 2.9.1+cu128 and CUDA 12.8, with the official dataset and tokenizer in place.

```
# 1. single-seed smoke test (recommended first)
python lenovo_C2G_train_gpt.py --seed 42 --out dir_out

# 2. full three-seed run (produces the submitted three-seed mean)
python lenovo_C2G_train_gpt.py --seeds 0,42,1234 --out dir_out

# 3. quantization + evaluation only (reuse a checkpoint, rerun TTT)
python lenovo_C2G_train_gpt.py --seed 42 --out dir_out --quantize_only --ttt
```

After a successful run, two bookkeeping steps promote this entry from a plan to a record:

(a) copy val\_bpb, val\_bpb\_std, seed\_results, artifact\_bytes and the three timing compliance flags from dir\_out/metrics.json into lenovo\_C2G\_submission.json and set verification\_status to "verified"; (b) replace the [TARGET] lines in lenovo\_C2G\_logs/\*.log with the real per-step output of the corresponding seed.

## 9. Attribution

This submission does not claim originality for any of the components listed below. Each is attributed to the public contribution that introduced it; the modifications made here are documented in `lenovo_C2G_■■■■.md`.

| Component                                | Attribution                                     | Change made here                               |
|------------------------------------------|-------------------------------------------------|------------------------------------------------|
| SP8192 tokenizer + GPTQ + SDClip         | @clarkkev (PR #1394)                            | embedding precision set to 8-bit at 20.0 sigma |
| Depth recurrence                         | @dexhunter (PR #1331, #1437)                    | none (aligned to SOTA settings)                |
| Parallel residuals                       | @Robby955 (PR #1412), @msisovic (PR #1204)      | none (start at layer 7)                        |
| Legal TTT framework                      | @abaybektursun (PR #549), @dexhunter (PR #1413) | none; four compliance switches held off        |
| QK gain 5.25                             | @bigbag (SOTA record 2026-04-09)                | none                                           |
| Hyperparameter set (EMA, WD, warmup)     | @X-Abhishek-X (PR #1445)                        | none; reused rather than re-searched           |
| Sliding-window evaluation                | @mattqlf (record 2026-03-19)                    | none                                           |
| Seed set 0 / 42 / 1234                   | this submission                                 | chosen for same-seed comparability             |
| Ablation matrix and 0.0006 decision rule | this submission                                 | designed here                                  |

## 10. Conclusion

The submitted recipe is a coherent, fully specified plan for reaching approximately 1.0810 BPB under a 600-second / 16 MB budget, assembled from components that are individually demonstrated in public records and combined in proportions that the records support. Its verification status is pending: the hardware required to execute it was not available on the machine that produced this package. The correct reading of this document is therefore that it specifies a falsifiable experiment with a precise success criterion (mean `val_bpb` below 1.0850 across seeds 0, 42 and 1234, artifact below 16,000,000 bytes, all nine compliance items satisfied), together with the exact procedure needed to test it.

## References

- [1] Official C2G parameter-golf repository, `records/track_10min_16mb*/submission.json` (primary source for all `val_bpb` and artifact byte values cited in this report).
- [2] Official baseline training script `train_gpt.py`, from which `lenovo_C2G_train_gpt.py` is derived.
- [3] Community pull requests #549, #1204, #1331, #1394, #1412, #1413, #1437, #1445 (component attribution, listed in Section 9).
- [4] `lenovo_C2G_brief.md`, the authoritative requirement document defining the 13 deliverables, the scoring rubric and the honesty requirement.

End of report. Document generated 2026-10-04. Verification status: PENDING\_VERIFICATION. This report must not be cited as evidence of a measured result until the reproduction procedure in Section 8 has been executed on the reference hardware.

